

Lakshya JEE 2.0 2025

Mathematics

Application of Derivatives

DPP: 1

- Q1** A particle moves in a straight line so that it covered a distance $at^3 + bt + 5$ metre in t seconds. If its acceleration after 4 seconds is 48 metre/(sec)², then a is equal to:
 (A) 1 (B) 2
 (C) 3 (D) 4
- Q2** An edge of a variable cube is increasing at the rate of 10 cm/sec. How fast the volume of the cube will increase when the edge is 5 cm long
 (A) 750 cm³/sec
 (B) 75 cm³/sec
 (C) 300 cm³/sec
 (D) 150 cm³/sec
- Q3** The point on the curve $6y = x^3 + 2$ at which y -co-ordinate is changing 8 times as fast as x -co-ordinate is
 (A) [4, 11]
 (B) [4, -11]
 (C) [-4, 11]
 (D) [-4, -1]
- Q4** The equation of a motion of a stone, thrown vertically upward is $s = ut - 6.3t^2$, where the unit of s and t are cm and sec. If the stone reaches at maximum height in 3 sec. then $u =$
 (A) 18.9 cm/sec
 (B) 12.6 cm/sec
 (C) 37.8 cm/sec
 (D) None of these
- Q5** A stone is dropped into a quiet lake and waves moves in circles at a speed of 4 cm/sec. At the instant, when the radius of the circular wave is 10 cm. How fast the enclosed area is increasing.
 (A) 40π (B) 80π
 (C) 100π (D) 60π
- Q6** A particle starts moving from rest from a fixed point in a fixed direction. The distance s from the fixed point at a time t is given by $s = t^2 + at - b + 17$, where a, b are real numbers. If the particle comes to rest after 5 sec at a distance of $s = 25$ units from the fixed point, then values of a and b are respectively:
 (A) 10, -33
 (B) -10, -33
 (C) -8, -33
 (D) -10, 33
- Q7** The coordinates of the point on the ellipse $16x^2 + 9y^2 = 400$ where the ordinate decreases at the same rate at which the abscissa increases, are:
 (A) (3, 16/3)
 (B) (-3, 16/3)
 (C) (3, -16/3)
 (D) (3, -3)
- Q8** Coffee is draining from a conical filter, height and diameter both 15 cm into a cylindrical coffee pot diameter 15 cm. The rate at which coffee drains from the filter into the pot is 100 cm³/min. The rate in cm/min at which the level in the pot is rising at the instant when the coffee in the pot is 10 cm, is:
 (A) $\frac{9}{16\pi}$



- (B) $\frac{25}{9\pi}$
 (C) $\frac{5}{3\pi}$
 (D) $\frac{16}{9\pi}$

Q9 The surface area of a cube is increasing at the rate of $2 \text{ cm}^2/\text{sec}$. When its edge is 90 cm , the volume is increasing at the rate of:

- (A) $1620 \text{ cm}^3/\text{sec}$
 (B) $810 \text{ cm}^3/\text{sec}$
 (C) $405 \text{ cm}^3/\text{sec}$
 (D) $45 \text{ cm}^3/\text{sec}$

Q10 A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa is

- (A) $(2, 6)$
 (B) $(2, -6)$
 (C) $(\frac{9}{8}, -\frac{9}{2})$
 (D) $(\frac{9}{8}, \frac{9}{2})$

Q11 The rate of change of the volume of a sphere w.r.t. its surface area, when the radius is 2 cm , is

- (A) 1 (B) 2
 (C) 3 (D) 4

Q12 A cube of ice melts without changing its shape at the uniform rate of $4 \text{ cm}^3/\text{min}$. The rate of change of the surface area of the cube, in cm^2/min , when the volume of the cube is 125 cm^3 , is

- (A) -4 (B) $-16/5$
 (C) $-16/6$ (D) $-8/15$

Q13 The radius of a right circular cylinder increases at the rate of $0.1 \text{ cm}/\text{minute}$, and the height decrease at the rate of $0.2 \text{ cm}/\text{minute}$. The rate of change of the volume of the cylinder, in $\text{cm}^3/\text{minute}$, when the radius is 2 cm and the height is 3 cm is

- (A) -2π
 (B) $-\frac{8\pi}{5}$

- (C) $-\frac{3\pi}{5}$
 (D) $\frac{2\pi}{5}$

Q14 Water runs into an inverted conical tent at the rate of $20 \text{ cu ft}/\text{min}$ and leaks out at the rate of $5 \text{ cu ft}/\text{min}$. The height of the water is three times the radius of the water's surface. The radius of the water surface is increasing when the radius is 5 ft , is

- (A) $\frac{1}{5\pi} \text{ ft}/\text{min}$
 (B) $\frac{1}{10\pi} \text{ ft}/\text{min}$
 (C) $\frac{1}{15\pi} \text{ ft}/\text{min}$
 (D) None of these

Q15 The radius of a right circular cylinder increases at a constant rate. Its altitude is a linear function of the radius and increases three times as fast as radius. When the radius is 1 cm the altitude is 6 cm . When the radius is 6 cm , the volume is increasing at the rate of $1 \text{ cm}^3/\text{s}$. When the radius is 36 cm , the volume is increasing at a rate of $n \text{ cm}^3/\text{s}$. The value of n is equal to

- (A) 12 (B) 22
 (C) 30 (D) 33

Q16 The distance travelled s (in *metre*) by a particle in t seconds is given by, $s = t^3 + 2t^2 + t$. The speed of the particle after 1 second will be

- (A) $8 \text{ cm}/\text{sec}$
 (B) $6 \text{ cm}/\text{sec}$
 (C) $2 \text{ cm}/\text{sec}$
 (D) None of these

Q17 A particle moves along the curve $y = x^{\frac{3}{2}}$ in the first quadrant in such a way that its distance from the origin increases at the rate of 11 units per second. The value of $\frac{dx}{dt}$ where $x = 3$ is, t is time in seconds.

- (A) 4
 (B) $\frac{9}{2}$



(C) $\frac{3\sqrt{3}}{2}$

(D) 12

Q18 A horse runs along a circle with speed of 20 km/h. A lantern is at the centre of the circle. A fence is there along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $\frac{1}{8}$ of the circle (in km/h) is

(A) 20

(B) 40

(C) 30

(D) 60

Q19 A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is

(A) $\frac{1}{54\pi} \text{ cm/min}$

(B) $\frac{5}{6\pi} \text{ cm/min}$

(C) $\frac{1}{36\pi} \text{ cm/min}$

(D) $\frac{1}{18\pi} \text{ cm/min}$



Answer Key

Q1 (B)

Q2 (A)

Q3 (A)

Q4 (C)

Q5 (B)

Q6 (B)

Q7 (A)

Q8 (D)

Q9 (D)

Q10 (D)

Q11 (A)

Q12 (B)

Q13 (D)

Q14 (A)

Q15 (D)

Q16 (A)

Q17 (A)

Q18 (B)

Q19 (D)



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